

Preliminary Design Considerations of an Initial Concept of a Wearable Device for Elderly People with Dementia

Keywords: Wearable devices, Dementia, Product design, Monitoring

Introduction & Objectives

One of the most significant socio-demographic changes of the 21st century is the progressive ageing of the population, which is one of the main challenges facing society today. This change has repercussions in various areas, including health. The economic, cultural and social and family demands, the increase in the number of elderly people, there has been a change in the priorities of research, with a particular focus on pathologies associated with ageing. Dementia is a chronic disease of the brain, characterised by a progressive decline in cognitive, motor and behavioural levels, compromising the autonomy of these individuals and making it difficult to carry out activities of daily living. The main goal of this research is to develop a wearable device that allows continuous and non-invasive monitoring of the activities and routine patterns of the elderly, facilitating the monitoring of relevant changes in the pathology of the elderly.

Methodology

This work adopted an iterative, user-centred approach to the development of a wearable device. To ensure a generally user-centred design, both functional and non-functional requirements were studied and established based on the findings presented in the scientific literature. Following the iterative design for Elderly People with Dementia, a focus on the functional requirements addressed the essential technical features of the device, including non-invasive location tracking, sleep pattern monitoring, fall detection, HRV and functionality, heart rate monitoring and blood oxygen saturation measurement.

The non-functional requirements, focused on user interaction aspects, divided into four domains:

- Physical Domain Requirements, related to ergonomics and ease of use;
- Functional Domain Requirements, addressing user acceptance and emotional and being with the device;
- Cognitive Domain Requirements, focus on intuitive interactions and usability;
- Successful Implementation Requirements, involves integration of real users in real world applicability.

These non-functional domains are visually represented and detailed in the circular diagram shown in Figure 1 offering a clearer overview of these interactions.



Figure 1: Circular diagram of design requirements for wearable devices (Pfeiffer et al., 2019, p. 25)

Development

The functional requirements highlight the core capabilities that the device must provide, encompassing a comprehensive set of features for continuous and effective monitoring. In parallel, the non-functional requirements address dimensions that influence the quality of user interaction, including user acceptance, physical ergonomics, emotional perception, and overall usability. These aspects are fundamental to providing an empathic relationship between the user and the device, thereby enhancing acceptance, comfort, and long-term adherence.

Within this framework, the initial was identified as the optimal anatomical location for device placement. This choice is supported by its minimal requirements, ease of use, and strategic relevance for physiological monitoring, given its proximity to peripheral vessels and the relatively stable nature of arteriole pulsations. To define the physical form of the device, an analysis was conducted of the various form factors available on the commercial market. This assessment considered both qualitative and quantitative aspects of existing solutions, including morphological, structural, and technological aspects. It is particularly important to emphasize that, since the elderly with dementia are the main users, the design of the device must be placed in its accessibility through physical elements. Regarding the size of the device's screen, the relationship between it and the anthropometry of the wrist must be in line with the design. In this way, different sizes are recommended for men and women due to their different anthropometric measurements, and the screen will only be used for viewing and not for interaction, only a model that can serve as a form factor as a wearable will be developed. Since one of the functional requirements is to monitor an element that can give feedback urgently, a study of cognitive profiles and sources of interaction factors was carried out. Since attention was paid to the fact that it should be easily visible and distinctive for the type of action it controls, given that these are areas with some cognitive difficulties.

In the same, ergonomic studies were carried out at a superficial level, noting that the problem with the greatest potential to current solutions would be in the center of the device, allowing both users who wear the device on their right wrist and those who wear it on their left wrist. In addition, the design of the buttons has been thought out for the type of action it performs and the way it is activated. In the case of the emergency button, it has been given the role of a large, visible, and distinctive button, and the correct shape so that it can be easily pressed. The other buttons, on the other hand, were given a more subtle and integrated design so that it would be easily pressed by the thumb, ensuring a precise and accurate touch.

Also, as part of the ergonomic studies, it was analyzed the ergonomics of the wrist and how the device could be made more comfortable for long use, given that these are highly sensitive areas. Thus, while wearing devices, going for device a more adjusted shape for long use with adherence and placement on the wrist, but also with accurate and accurate measurement of vital data.

With all this, we moved on to these dimensional representation phase, which offers a more real and detailed visualization of the form as they can be used or allow us to study the internal arrangement to accommodate the sensors. This visualization also included a study of button shapes to analyze how they fit into the continuity of the device's overall design.

All the considerations described above are visually represented by a flowchart in Figure 2.

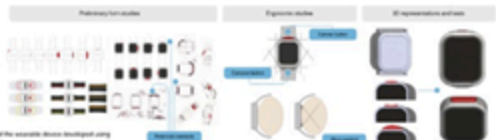


Figure 2: Preliminary objectives of the wearable device developed using center design software, Adobe Photoshop

Results

From preliminary design studies, the rectangular shape emerged as the most advantageous for users with dementia, addressing both cognitive and ergonomic needs. Its compact and familiar form facilitates adoption, promoting user confidence.

Compared to other geometries:

- Circular devices provide a conventional and intuitive appearance that fits better placement than to current constraints.
- Due to cylindrical devices offer complexity both visually and tactile functions that may be unfamiliar to elderly users, potentially complicating adoption and reducing future usage.

In contrast, the rectangular shape offers flexible layout placement for intuitive interaction while maintaining a simple form of greater cognitive accessibility, and an intuitive approach that considers device physics and negative effects. Beyond the shape, sensor size plays a critical role. Between two commercial sizes (40mm and 50mm), the latter sensors better serve fit and comfort, particularly for female users, while maintaining sufficient display area for essential information. Given that the device will not act as a touch interface, a larger screen is unnecessary, further supporting the circular form.

Following these considerations, a series of digital sketches were developed to visually translate the specifications into initial design concepts aligned with the cognitive and physical needs of elderly individuals with dementia.

Finally, to validate the concept, prototype development – including its electronic components – is essential. User testing with the target population will provide valuable feedback for further refinement and validation.

Conclusions

This study aimed to identify the most appropriate design characteristics for a wearable device intended for elderly with dementia, considering both specific needs and functional requirements. Through a comparative analysis of geometric shapes, the square or rectangular form was determined to be the most advantageous, due to its conventional appearance, simplicity, and ergonomic potential.

In parallel, an evaluation of display sizes concluded that the smallest format offers the best balance between visual clarity and physical comfort, especially considering the device's touch interaction and the need for a compact and friendly design.

Beyond of these findings, a framework of preliminary considerations was developed to visually translate the selected design process into a functional concept. These visual representations serve as a foundational step towards understanding physical and cognitive requirements of the intended design for wearable technology for cognitively impaired populations.

The results contribute to the development of more accessible and inclusive wearable solutions. Future work will encompass a user-centered approach by developing a functional model and conducting usability testing with real users, ensuring the proposed solution meets the cognitive, physical, and emotional needs of the target population.



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